The 2025 Imsil earthquake swarm, southwestern Korea

Detection, relocation, template matching and earlier activity on the source fault

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Abstract. On 2 November 2025 (UTC), a swarm of small earthquakes occurred at mid-crustal depth beneath Imsil, southwestern Korea. The Korea Meteorological Administration (KMA) listed 24 events of local magnitude M_L 1.0–1.9 within 4.5 h. We rebuilt the catalog with the detection, relocation and template-matching pipeline adopted for the Haenam sequences, changing only the region values. Every change of setting was compared before adoption and recorded. Template matching with the 28 best-located events finds 235 events, 207 of them new. All but one fall within 5 h on 2–3 November, and the one exception follows at 04:23:32 on 3 November. The 101 events whose 95 % location half-widths are at most 25 m outline a near-vertical plane striking 268.7° and dipping 88.2°. Their root-mean-square (RMS) distance from the plane is 7.4 m, and they span about 241 m along strike and 235 m in depth. The swarm started low in the west and spread east and up the plane. A diffusion front fitted to the outermost events gives a hydraulic diffusivity of 0.50–0.63 m^2/s, depending on the length of the fit window. A template-matching scan of 3,198 days from 2018 to 2026 finds three earlier events on 15 April 2019. Their amplitudes are 0.024–0.043 times that of the M_L 1.9 event, and KMA did not catalog them. Relocated together with the 2025 swarm, they lie 266–305 m south of the 2025 plane, at the same depths and within its along-strike extent. They therefore point to a second, parallel fault strand that was active six years before the swarm. Compared with Park et al. (2026), who studied the same swarm, our catalog holds 47% of their events. The difference is in events below their M_w 0.8, and it follows from their larger network, wider frequency band and lower detection threshold.

1 The swarm and the data

The swarm occurred on 2–3 November 2025 between 20:05 and 00:35 UTC (05:05–09:35 Korean time on 3 November). Its largest event was the M_L 1.9 at 20:38:35 UTC. Before this study, the swarm had been analyzed only with an earlier, quicker pipeline: hand picks on the M_L 1.9, deep-learning detection at 11 stations over three days, and a double-difference relocation of 18 events.

We used the continuous records of the 54 KMA stations (network code KS) within 100 km of the swarm (Figure 1), from 1 September 2025 to 31 January 2026 (153 days). No data of the Korea Institute of Geoscience and Mineral Resources (KIGAM) exist for 2025, and the nearest Chonnam National University stations started in September 2026, so the network is KS only. The nearest station is IMSB (6.9 km). Template matching used the 22 stations within 59 km.

2 Methods

The pipeline is the one adopted for the 2020–2026 Haenam sequences and the 2024 Buan sequence (Seo et al., 2025). Its code was copied unchanged: the picking, association, location, waveform export, magnitude and template-matching scan modules are byte-identical to the Haenam code, as checked by checksum. Only the region values (association box, relocation box, station table, period) were changed, each by a stated rule. Each choice that the Haenam rules did not settle was given evidence in a numbered notebook and recorded in a decision log (Section 5). Table 1 lists the settings.

Detection and location. A deep-learning phase picker of the PhaseNet family (Zhu and Beroza, 2019) was applied at 100 samples per second, keeping picks with probability ≥ 0.2. Picks were grouped into events (*association*) with PyOcto (Münchmeyer, 2024) within 34.85–36.25° N, 126.50–128.10° E and 0–30 km depth. An event required at least 6 picks, among them 3 P and 3 S picks, and 2 stations with both. Events were located with HYPOINVERSE (Klein, 2002) in the layered model of Kim and Kim (1983). On the swarm days, the picker found all 24 KMA events. The M_L 1.9 located 0.11 km from its hand-pick epicenter at 14.6 km depth. The median differences between automatic and hand picks were 0.01 s for P and 0.032 s for S.

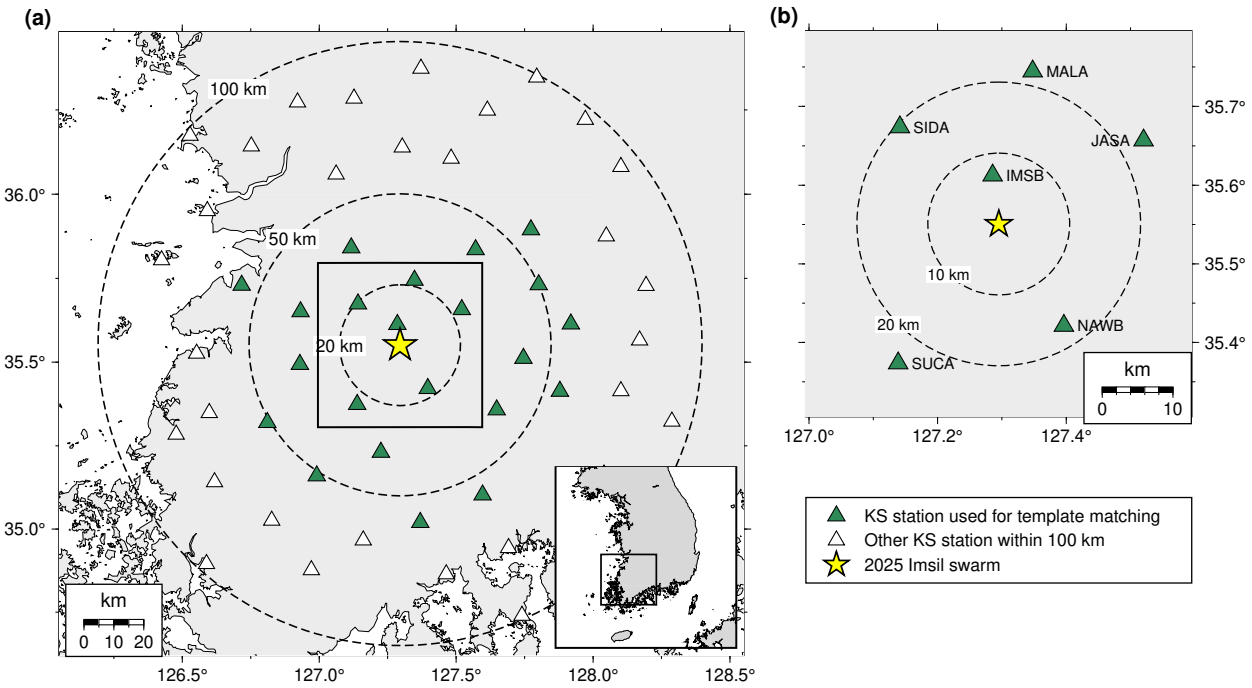


Figure 1. Stations. (a) The 54 KS stations within 100 km of the swarm (star), with the 22 stations used for template matching in green; dashed circles at 20, 50 and 100 km. Inset: location in Korea. (b) The stations within the box of panel (a), with names.

Relocation. Events in the Imsil box that passed a location-quality gate were relocated by the double-difference method (Waldhauser and Ellsworth, 2000), which solves for the relative positions of nearby events. The gate required an azimuthal gap below 180°, more than 7 phases, an RMS residual below 1 s, errors below 5 km and a depth of at least 10 km. Differential times came from the picks and from waveform cross-correlation: 5–20 Hz, windows of $\pm$0.5 s, interpolation to 1,000 Hz, correlation coefficient (CC) ≥ 0.7, and at least 4 observations per event pair. Uncertainties are the 2.5–97.5 % ranges of 1,000 bootstrap replicas, each with the data resampled. The *reference catalog* keeps the events whose 95 % half-widths are at most 25 m in east, north and depth separately. Planes were fitted by singular-value decomposition (SVD), a total-least-squares fit to the hypocenters.

Magnitudes. Local magnitudes come from the scale of Sheen et al. (2018), computed as in Haenam. For relative size we also use the ratio of 5–20 Hz S-wave peak displacement to that of the M_{L} 1.9 event, recorded on the same channel so that the instrument response cancels. Each event's ratio is the median over the channels on which the M_{L} 1.9 has a signal-to-noise ratio of at least 300. The scale of Park et al. (2026) converts this ratio to their M_{w}.

Template matching. The 28 events of the reference catalog served as templates for a network matched filter (Beaucé et al., 2018). The filter used 5–20 Hz, a 1.5 s P window on the vertical and 2.0 s S windows, each starting 0.5 s before the pick, at 22 stations; S was taken on the vertical at NAWB, KWJ2, whose horizontals were re-instrumented during the period. A detection must exceed 10 times the median absolute deviation (MAD) of the network correlation, computed per day and per station configuration, with at least 4 stations; hits closer than 4 s are merged. Two settings differ from Haenam. First, every day trace is shifted onto the exact 10 ms sample grid before templates are cut. Without this shift, templates at stations sampling off the grid were compared with the wrong sample, and their self-detection CC was 0.841 instead of 1.000. Second, the threshold is 10 × MAD with no absolute CC floor, instead of Haenam's 12 × MAD with a CC floor of 0.20. The catalog sizes for the four combinations were 187, 194, 210 and 235.

Joint relocation of detections. Each new detection inherits its template's picks, shifted to the detection origin; the picks only place the correlation windows. Detections are tied to the 28 reference events and to each other by cross-correlation times alone, with the relocation settings above, the bootstrap and the 25 m rule.

Table 1. Pipeline and settings. The detection and relocation settings equal Haenam's; the two template-matching changes are explained in the text.

Step	Settings
Stations and period	54 KS stations within 100 km; 1 Sep 2025 – 31 Jan 2026 (153 days)
Picking	PhaseNet family, 100 Hz, probability ≥ 0.2
Association	PyOcto; ≥ 6 picks, ≥ 3 P, ≥ 3 S, 2 stations with both
Location	HYPOINVERSE, Kim & Kim (1983) model
Relocation	double difference; picks + cross-correlation (5–20 Hz, CC ≥ 0.7); 1,000 bootstrap replicas; reference = 95 % half-widths ≤ 25 m
Local magnitude	Sheen et al. (2018) scale, Haenam implementation
Template matching	28 templates, 22 stations, 5–20 Hz; P 1.5 s, S 2.0 s; 10 × MAD, no CC floor, ≥ 4 stations; grid-aligned traces
Long scan	same templates, 2018-01-01 to 2026-10-03 (3,198 days)

Long scan. The same templates were run over every day from 1 January 2018 to 3 October 2026, 3,198 days. The vertical was used for S at the 9 stations whose response changed during these years. KS.IMSB, the nearest station, has records from 2019-03-18.

3 Results

3.1 Deep-learning catalog and the reference events

The detection run located 1,695 events in and around the region from 484,833 P and 638,314 S picks. It recovered 24 of 24 KMA-listed Imsil events and 86 of 89 KMA events in the association box. The Imsil box holds 125 events: 31 before the swarm, 39 during it and 55 after it. Apart from the swarm, the box contains mainly shallow daytime events, probably quarry blasts, which the depth floor removes (46 of the well-located ones).

Of the 34 relocated events, 28 meet the 25 m rule, 22 of 24 KMA-listed events among them. Their median 95 % half-widths are 1.9, 1.3 and 2.1 m in east, north and depth. They occupy 126 m east–west, 27 m north–south and 186 m in depth, on a plane of 272° / 87° (strike/dip). The earlier relocation of 18 events agrees after one rigid shift of the whole cluster: the median residuals of the 17 matched events are 1, 1, 2 m in east, north and depth.

3.2 Magnitudes

Only 8 of 28 reference events have a local magnitude from at least three stations with a pick and a signal-to-noise ratio of at least 2. Most swarm events are too small for the KS network at its usual distances. Where both exist, our M_L minus the KMA M_L has a median of −0.13 and an RMS of 0.17. Three stations have large residuals: NAWB +0.54, KWJ2 −0.55 and EURB −0.46. These stations carry response epochs that differ between the two metadata sources. With the KMA response files instead of the master metadata, the NAWB and KWJ2 residuals become +0.04 and +0.15, and 14 events get a magnitude. Which source to use is still open (D4). The Haenam relative-magnitude method needs anchor events with at least 20 picked stations; Imsil has 0, so that method does not transfer. For this reason the timeline uses amplitude ratios (Figure 2a).

3.3 Template matching

Template matching finds 235 events: the 28 templates and 207 new detections (Figure 2). 234 fall between 19:57:38 and 00:53:41 UTC, and the last follows alone at 04:23:32 on 3 November. No other event in the five-month period matches the templates. Template 400005 (event 20251102202656), a small event that KMA did not list, accounts for the most detections (51). Of the new detections, 114 are isolated and 93 lie within 15 s of a stronger detection. A template could also match the later phases of a larger event at a fixed lag, but the observed lags show no excess over a randomized control. All detections therefore went to relocation, where the correlation links and the bootstrap act as an independent check.

The detections follow the four phases P1–P4 defined by Park et al. (2026) (Figure 2a). For 140 detections

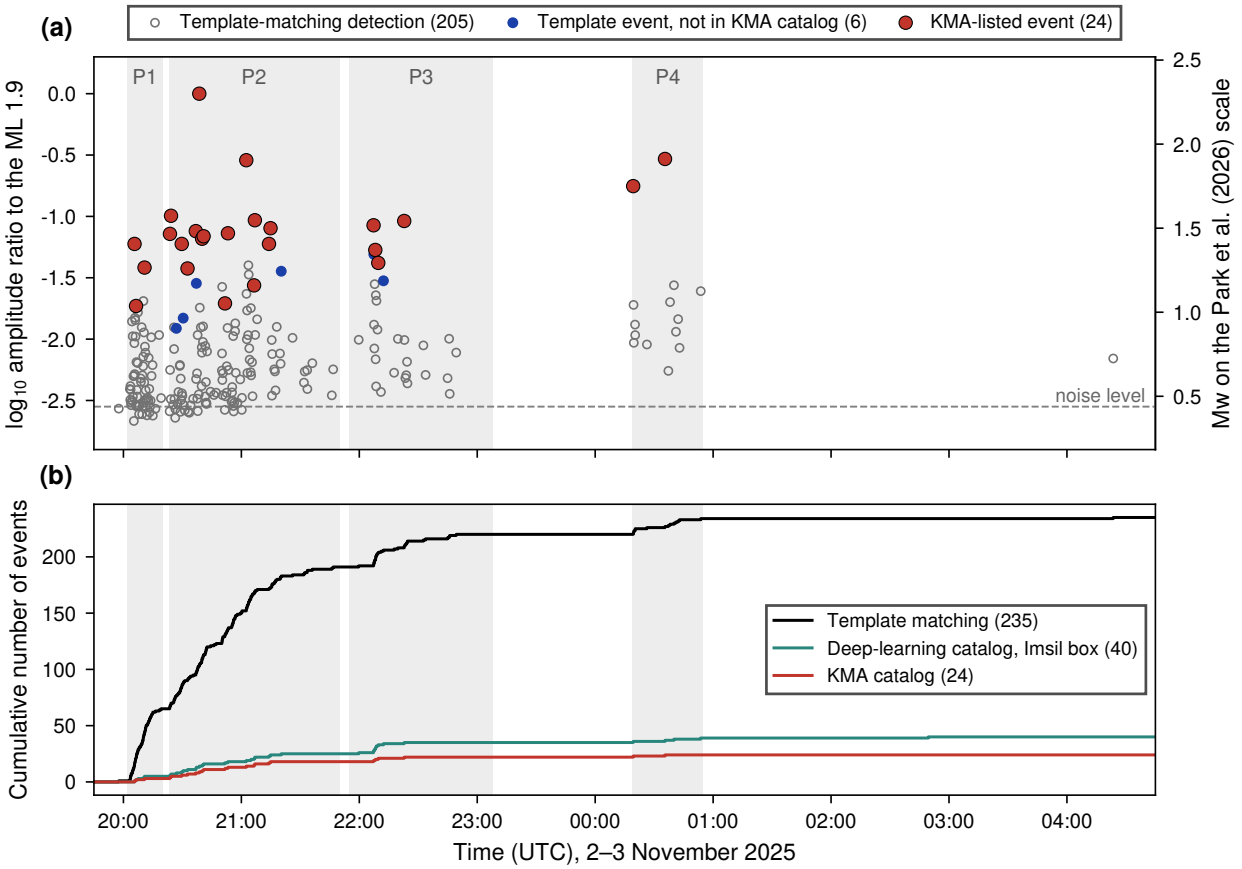


Figure 2. The swarm in time. (a) Size of the 235 template-matching events: the 5–20 Hz S-wave peak-displacement ratio to the M_L 1.9 (median over the channels on which the M_L 1.9 has a signal-to-noise ratio of at least 300). The right axis converts the ratio to M_w with the scale of Park et al. (2026). Red: KMA-listed; blue: templates not listed by KMA; open: new detections. The dashed line is the median reading of the same measurement before the P arrival (noise). Gray bands: phases P1–P4 of Park et al. (2026). (b) Cumulative numbers of events: template matching, the deep-learning catalog (Imsil box, 19:30–05:00 UTC) and KMA.

the amplitude ratio is within a factor of 2 of the same measurement on the noise before P, so their sizes are upper bounds. In the swarm window, the deep-learning catalog has 40 events in the Imsil box and KMA lists 24 (Figure 2b).

3.4 Geometry

Of the 235 events, 212 were relocated (184 detections), and 101 meet the 25 m rule: the 28 templates and 73 detections (Figure 3). Their median 95 % half-widths are 1.8, 1.3, 1.2 m (templates) and 16.6, 17.1, 75.9 m (detections) in east, north and depth. The SVD plane strikes 268.7° and dips 88.2° to the north (95 % ranges 267–271° and 87–90°). The RMS distance from the plane is 7.4 m, and 99 % of the events lie within 15 m of it. The events span 210, 29, 209 m in east, north and depth (95 %). Absolute depths of a double-difference cluster are not resolved, because the cluster keeps the mean starting depth of its events. Two relocation runs agree to a few meters in relative position but differ by 520 m in absolute position. The absolute depth is best taken from the well-recorded single-event locations: 13.79 km.

3.5 Migration

The first event, at 19:57:38, lies at the lower western end of the patch, and later events spread east and up (Figure 3b, Figure 4b). If a pressure front spreading from the first event triggered the swarm, the outermost events should follow $r^2 = 4\pi Dt$, where D is the hydraulic diffusivity (Shapiro et al., 1997). Fits through the front-setting events give D = 0.50, 0.59 and 0.63 m^2/s for fit windows of 0.5, 1 and 2 h. Their bootstrap 95 % ranges are 0.42–0.74, 0.42–0.67 and 0.48–0.69 m^2/s (Figure 4a). We also fitted a back front, the inner edge of a quiet zone that opens after the pressure source stops (Parotidis et al., 2004). It is not stable: its

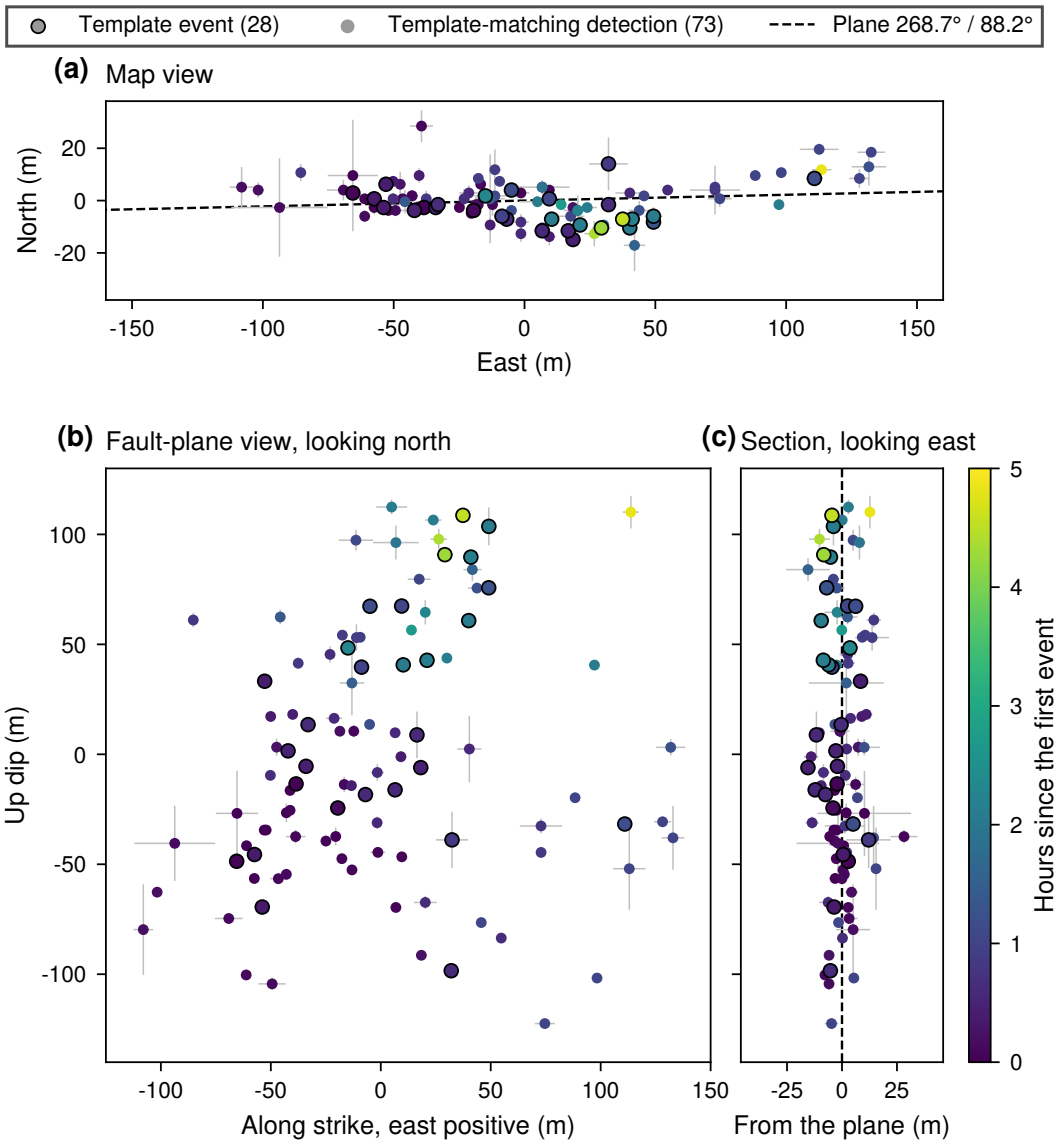


Figure 3. Geometry of the 101 events that meet the 25 m rule, colored by time; templates have black edges. Gray bars are 95 % half-widths. (a) Map view with the trace of the fitted plane. (b) The plane seen from the south (along strike against up dip). (c) Section along the plane: distance from the plane against up dip. Positions are relative to the centroid.

start time ranges over 1.0–25.9 min in resamples, so it is not interpreted. The choice of time scale is still open (D24).

3.6 Comparison with Park et al. (2026)

Park et al. (2026) studied the same swarm with one template (the M_L 1.9), 60 stations, a 10–40 Hz band and visual vetting. Their per-event catalog is not public, so their side of each comparison is the number printed in their paper (Table 2). We reproduce their magnitude scale to 0.11 RMS on the events whose magnitudes or amplitude ratios they quote. Of the 263 events they have and we lack, 71–95% are below their M_w 0.8. Their mechanism of the M_L 1.9 predicts the ranking of our station P/S amplitude ratios (Spearman correlation 0.59, $p = 0.004$). IMSB lies near a P-wave node, which is why its P is weak. They read the fall of waveform similarity during the swarm as a change of the source with time. In our data, the fall is carried by the distance between events ($t = -14$ / -22 at MALA and NAWB) rather than by time ($t = -2.1$ / $+2.8$), that is, by the migration. Their b-value of 1.40 ± 0.07 rests on a magnitude scale with slope 0.73 borrowed from another region. On their scale our b-value is 1.33 ± 0.18, and the same data give 1.53–1.02 for slopes from 2/3 to 1. A “high b-value” is therefore partly a statement about the borrowed slope.

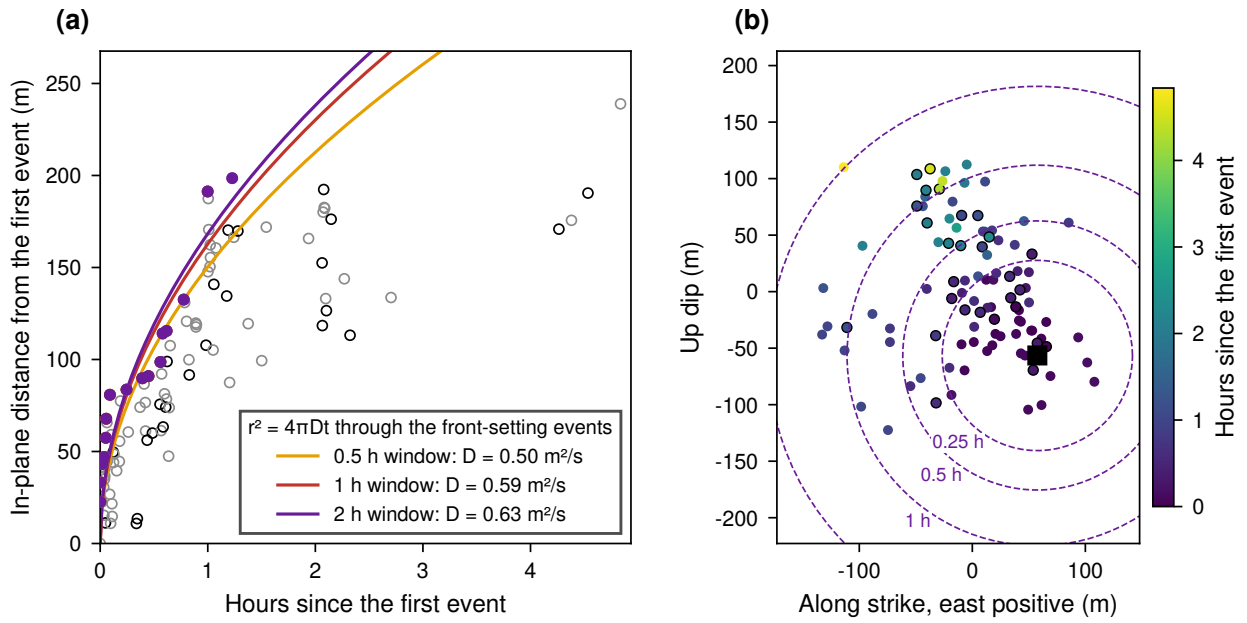


Figure 4. Migration. (a) In-plane distance from the first event against time; filled symbols are the front-setting events of each fit window, and the curves are $r^2 = 4\pi Dt$ fits for windows of 0.5, 1 and 2 h. Black-edged: templates. (b) The events on the plane, colored by time; the square is the first event, and the dashed circles are the 2-h front at 0.25, 0.5, 1 and 2 h.

Table 2. This study and Park et al. (2026).

Topic	Park et al. (2026)	This study
Catalog	498 events; 60 stations; 10–40 Hz	235 events (47%); 22 stations; 5–20 Hz
Share per phase	–	P1 47%, P2 44%, P3 55%, P4 64%
Depth	13.6–14.0 km	13.79 km (single-event locations)
Plane	89° / 89°	268.7° / 88.2°
Extent, along strike × depth	300 × 350 m	237 × 225 m; 37 m thick (99 %)
Diffusivity, their method	0.45–0.61 m^2/s	0.34–0.40 m^2/s
b-value	1.40 ± 0.07	1.33 ± 0.18 on their scale

3.7 The long scan, 2018–2026

The long scan returned 6,369 detections (Figure 5):

- 212 in the 2025 swarm window, the control, which reproduces the swarm with the changed S components;
- 6,135 on 6–8 March 2018, which are noise triggers: a median stack of the 400 strongest shows no arrivals at any station;
- 22 others, in three groups.

The 12 detections of group B come from template 400005 alone, between September 2023 and March 2025. Their signal is seen only at JASA, where the stacked envelope rises to 4.6 times the pre-event level (2.0 at IMSB). Their similarity to each other is low (0.39 (0.25–0.76)). JASA carries 0.182 of this template's weight, more than for any other template. Group B is therefore a repeating local source near JASA, not activity on the Imsil fault. KMA lists 23 events within 20 km of JASA in 2023–2025, and none of them coincides with a group-B detection. The 7 single-template detections of group C show no arrivals. Group A consists of three events on 15 April 2019, each detected by several templates.

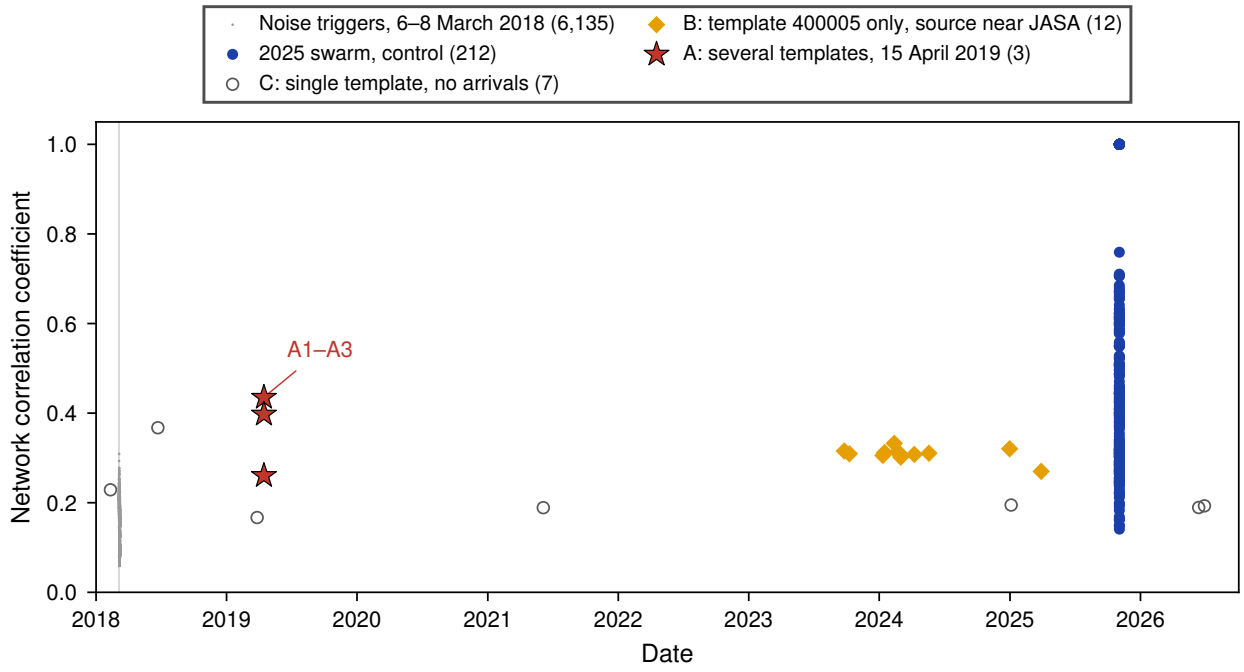


Figure 5. Template-matching detections from 1 January 2018 to 3 October 2026 (3,198 days), by network correlation coefficient. Gray band and dots: the noise burst of 6–8 March 2018.

3.8 The events of 15 April 2019

A1 (14:17:35 UTC), A2 (14:17:39) and A3 (14:43:15) were detected by 3, 7 and 6 templates. Their waveforms repeat those of the 2025 templates (Figure 6). The correlation in the S window is 0.80–0.92 at the nearest station, IMSB, and 0.24–0.67 at JEU2, GKSA and MALA. It is at most 0.11 at SUCA and JLSA, where S is taken on the vertical and the stations were re-instrumented between 2019 and 2025. Their S amplitudes are 0.025, 0.043 and 0.024 times that of the M_L 1.9, about M_L 0.3–0.5 if M_L scales one-to-one with log amplitude. Of the 2025 swarm events with amplitude ratios of 0.016–0.032, 0.032–0.050 and larger, KMA cataloged 16% / 43% / 100%, so events of this size are usually missing from its catalog.

Relocated together with all 2025 events and with the same settings, A1–A3 lie 266–305 m south of the 2025 plane (Figure 7). For comparison, 99 % of the 25 m events of the same run lie within 16 m of the plane. Their 95 % half-widths are 20–35 m north–south, 83–130 m east–west and 36–67 m in depth. In the plane they lie +35 to +186 m along strike and −86 to −9 m up dip from the swarm centroid, inside the swarm's extent and at its depths (Figure 7c). Their correlation links come from 8–9 stations, because detections inherit the picks of their templates. At some stations the 2019-to-2025 links carry median residuals that point to timing or instrument changes: NAWB +93 ms, KCH2 −41 ms and JLSA −38 ms. Solved again against the fixed swarm without these stations, or with stable-epoch stations only, the events stay 256–328 m south.

Relocated alone, the three events cannot move as a group. Every equation compares one of them with another, so their centroid stays at its starting position, the location of their templates. The solo run therefore cannot test the offset. It does show how they lie relative to each other: within 38 m alone and 24 m in the re-solve, against 168 m in the joint run, whose along-strike spread is within its uncertainty (Figure 7d). A1–A3 are best read as a tight group 266–305 m south of the 2025 plane, on a structure parallel to it.

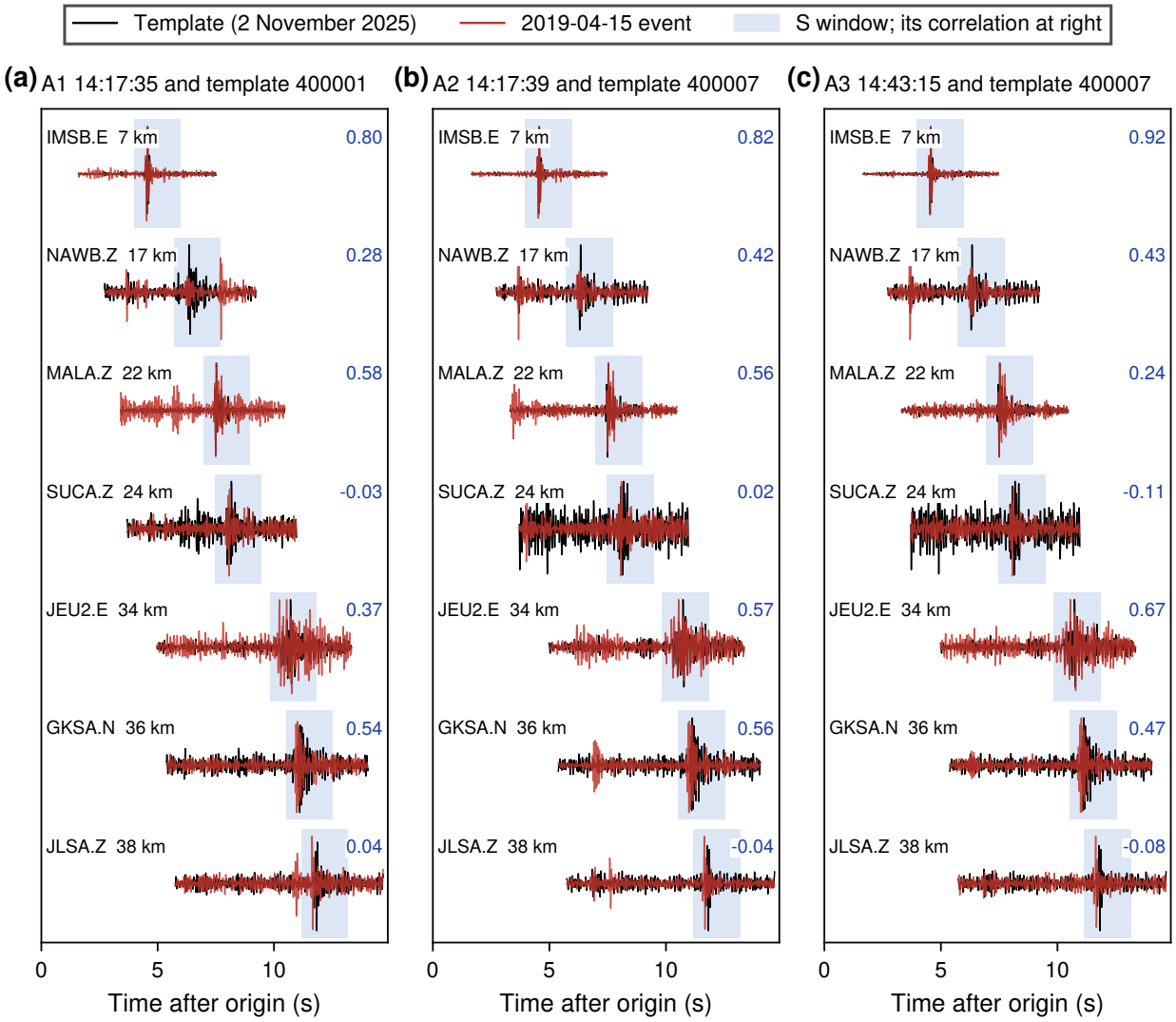


Figure 6. Waveforms of A1–A3 (red) against their templates (black) at the seven nearest stations with a template S pick, 5–20 Hz, each trace normalized. The S window (shaded) is the one the scan correlates; the number is its correlation at the detection time. S is shown on the horizontal component with the higher correlation, or on the vertical where the scan uses the vertical.

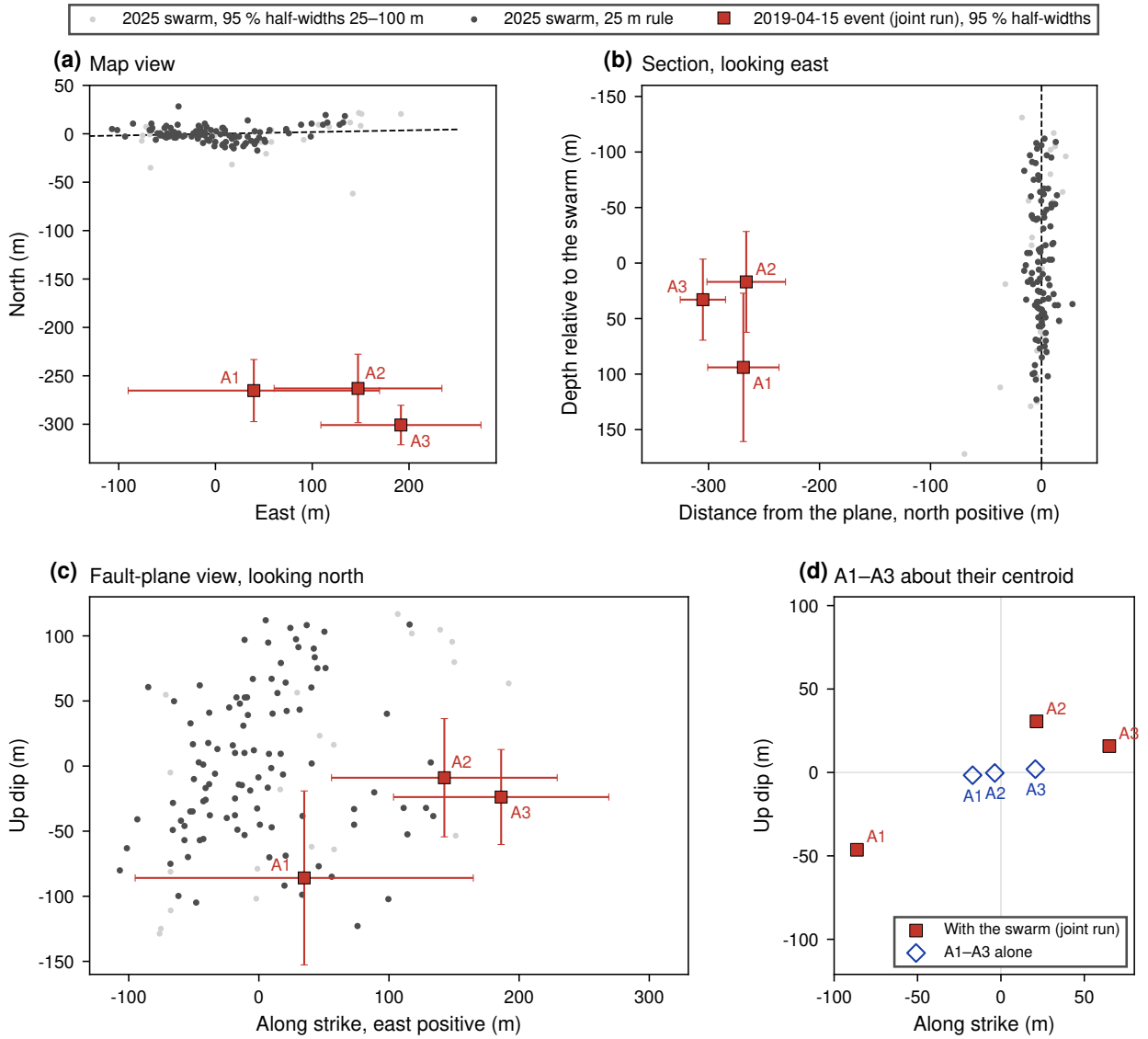


Figure 7. Position of A1–A3 relative to the 2025 swarm (joint relocation; error bars are 95 % half-widths). (a) Map view with the trace of the 2025 plane. (b) Section looking east: distance from the plane against depth relative to the swarm centroid. (c) Fault-plane view. (d) A1–A3 about their own centroid in the joint run and in the relocation of the three events alone.

4 Interpretation

The swarm activated a near-vertical patch about 241 m long on an east–west plane, starting at its lower western end and spreading up and east. The spreading follows a diffusion front with D = 0.50–0.63 m^2/s, consistent with a pressure front. The events of 15 April 2019 show that the source region was active six years earlier, on a structure 266–305 m south of the 2025 plane and parallel to it. The three events are small and not cataloged by KMA. The joint relocation, the robustness tests and the waveform similarity all support this reading. Two limits remain. The 2019 events are tied to the swarm at only 8–9 stations. Some of those stations changed timing or instruments between 2019 and 2025, although removing them does not change the result. An independent check, with picks of the 2019 events themselves or with S−P times that do not depend on station clocks, has not yet been made. No other detection between 2018 and 2026 is an event on the Imsil source.

5 Open decisions and limitations

The following settings are still open in the decision log (`docs/decisions.csv`); each has its evidence in the notebook named.

- **D4, response metadata for M_L** (notebooks 05, 05a): the master metadata or the KMA response files; the station residuals at NAWB, KWJ2 and EURB decide.
- **D18, magnitude validator**: the Haenam validator quarantines every KMA response file because of a cosmetic warning; it changes the number of magnitudes.
- **D5, depth floor of 10 km** and **D6, onset windows** (notebook 04): an onset window moves the M_L 1.9 by 6.2 m.
- **D14 and D16**: relative magnitudes (not feasible, no anchors) and station exclusions for M_L.
- **D15**: adoption of the SVD plane, which is well constrained (267–271°, 87–90°).
- **D24**: time scale of the migration fits.

Template matching cannot find events unlike the templates, and the long scan has fewer stations before 2019. The absolute location of the swarm rests on single-event locations; double-difference positions are relative.

6 Data and code

All processing is in numbered notebooks (00–16) and scripts in the project repository (`https://github.com/seismoseo/imsil-2025-swarm`), with the decision log and the final catalogs. The figures and every number of this report are produced by notebook 16. KMA waveforms are not redistributed.

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